



Investment decisions under uncertainty—A methodological review on forest science studies

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ABSTRACT

Several parametric and non-parametric approaches have been developed to value financial assets. Yet, financial valuation techniques have only slowly percolated into disciplines concerned with the management of ecosystems. Particularly in forest management, decision-makers often find themselves confronted with extremely long time horizons and severely uncertain information. This requires careful valuation approaches, which are often underrepresented or even completely lacking in forest management. This paper gives a comprehensive overview on techniques for financial decision-making under uncertainty and develops future research needs.

First, we analyse different approaches from the expected utility framework as well as option pricing models and robust optimisation techniques as possible approaches to make decisions on forest investments and giving a short review regarding forestry-related applications. Afterwards we discuss the suitability of the presented approaches to support decisions in forestry and conclude that robust optimisation techniques should be developed further, especially since erroneous financial data is likely to occur, as well as deviations from the assumption of normality. Currently, the maximization of financial robustness is probably the most adequate approach for many long-term decisions in forestry, such as selecting the optimum tree species composition. Further development of this approach appears possible and necessary.

Finally, we come to the conclusion that even though it is intuitively clear that many long-term decisions should consider uncertainty, adequate financial valuation is not sufficiently developed within forest science. In the case of Central Europe, this may be an effect of ecological research dominating in forest science. Consequently, an intensification of the analysis of uncertainty in forest decision-making is necessary.

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1. Introduction

Financial analysis of investments is especially difficult in forestry because of the long-term character of many decisions (e.g. tree species selection for plantations, silvicultural treatments, etc.). Positive net revenues often lie far in the future and are therefore severely uncertain, yet forest managers must decide today where to invest money. Due to the fact that the financial return and other beneficial effects of today's investments will often be received by coming generations, the sustainability paradigm plays an important role in forest management. It is clear that forest managers have to integrate uncertainty into their decision-making processes if their aim is to provide robust net revenues for both current and future generations. In order to achieve this aim of robust net revenues, diversification is generally a well-accepted strategy in financial management. However, approaches for the integration of uncertainty in financial decision-making have only slowly percolated into the

discipline of forest management and other disciplines concerned with the management of ecosystems.

Investment literature differentiates between investment projects and investment programmes under uncertainty (e.g. Götze et al., 2008). To cover a broader range of studies considering investment uncertainties, we focused on both kinds of decisions, as the presented approaches for investment projects can be and have been used for investment program decisions (e.g. tree species selection).

Our paper especially addresses the forest scientist and gives a broad introduction into the background of the probabilistic valuation techniques. Finally, we identify the need for further development of robust optimisation techniques to enhance the reliability of results, especially under the assumption of erroneous financial data and for the aspect of separation between uncertainty and ignorance.

1.1. Decision making under uncertainty

Consideration of uncertainty for decision-making has been investigated in detail for many fields of decision analysis. Hence, a multitude of models have been developed in order to take uncertainty into account. An overview of different models is given by e.g. Götze

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et al. (2008). At first, a differentiation between different types of uncertainties is useful: According to the established literature (Knight, 1921) we distinguish if the probabilities for the various outcomes of a decision alternative are known or not.¹ The latter case is often described as a situation of ignorance.

On the one hand, the models which assume ignorance of the relevant probabilities still assume that the decision maker knows the possible outcomes (or possible future states)—which is still information that might be hard to get in practice. Quite well known decision models for this situation are for example the *Maximin*-, *Maximax*-, *Hurwicz*-, *Laplace*-, *Savage-Niehans*- and *Krelle*-rule (Götze et al., 2008, pp. 263–265; Bamberg and Coenenberg, 2006, pp. 127–140), which differ with respect to the assumptions on the assumed preferences of the decision maker. On the other hand, a multitude of models has been developed, which assume, that besides the possible future states also the state probabilities are known or can be derived from available data. Well known decision rules for this case of risk are the *Bayes rule*, the $\mu - \sigma$ criterion or the *Bernoulli criterion* (Götze et al., 2008, pp. 265–268).

Depending on the definitions and classifications of uncertainty, the available information and the investigated decision task, different decision support applications have been developed. Zimmermann (2000) stated that modelling of uncertainty should always be done within the context of quality and quantity of available information and therefore there is no single uncertainty theory that could claim to be the solution for every kind of uncertain phenomena.

Kangas and Kangas (2004) provided a broad overview of different uncertainty theories (*Frequentist*- and *Bayesian probability theory*, *Evidence*-, *Fuzzy set*- and *Possibility theory*) and their applications related to the field of decision making in forestry and natural resource management. They pointed out that probability theory and Fuzzy set theory² are the most common approaches in decision support tools and that classical probability theory could prove to be adequate for almost every situation within timber production decision-making.

As our focus in this paper is on investment decisions in forestry or ecosystem management with conscious consideration of financial risk as a result of uncertainty, we selected from the probabilistic framework the expected utility approach as a starting point and three closely related approaches (stochastic dominance, downside risk measures and mean-variance analysis) for long term financial decisions. Furthermore, we considered option pricing models and robust optimisation models as two extending approaches. The estimation of probability distributions is an important step in risk analysis performed within the presented approaches and therefore we first want to introduce concisely the most common technique. After a short theoretical description of the considered approaches, we give an overview of existing forest science studies that apply the previously introduced approaches. As a matter of course, there are even more approaches from the probabilistic framework such as sensitivity analysis or the decision-tree method (Götze et al., 2008), which are not considered in this study for various reasons.³

¹ Other authors (e.g. Hirshleifer and Riley, 2002) do not differentiate uncertainty according to Knight (1921) but will also differentiate between “hard” and “soft” probabilities according to the probability concept developed by Savage (1954).

² Fuzzy set theory was developed by Zadeh (1965) and is used for decision making dealing with inexplicit definitions. The approach and applications in forestry decision making have been described by Kangas and Kangas (2004) and therefore they are not included in this study.

³ Sensitivity analysis is a common way to consider uncertainty (by changing input variables and evaluating the effects on target variables) in scientific studies, as well in forestry financial evaluation (e.g. Klemperer, 1996, p. 320; Cabbage et al., 2007). For reasons of straightforwardness we did not address sensitivity analysis specifically as an approach to consider uncertainty but the technique is also related to other approaches (*Stochastic dominance*, *Information gap decision theory*). The *Decision tree method* (e.g. Klemperer, 1996, pp. 326) is not very common in forestry investment analysis, except the use in connection with other approaches like *options pricing models* (e.g. Yoshimoto and Shoji, 1998; Yoshimoto, 2002).

1.2. Generation of probability distributions

Most of the presented approaches are based on comparisons of the estimated probability distributions of possible investment outcomes (financial return, net present value). A routinely used method to derive these probability distributions for decision making under uncertainty is the Monte Carlo Simulation technique (MCS), where repeated random samples are considered for distributions of possible results. Inputs must be defined (e.g. timber prices and their volatility) and combined in a way that allows for any interdependencies (e.g. possible correlations between timber prices of different species). Random fluctuations of uncertain input parameters are simulated and the results (e.g. financial outcomes) of a large number of repeated simulations can be used to derive the relevant probability distribution functions. Finally, output parameters like means and variances can be determined for evaluations with parametric approaches or non-parametric approaches, based on the whole distribution functions or parts of them, can be carried out.

MCS is broadly used for risk assessment in financial evaluations due to its ability to incorporate different uncertain inputs, but is also common in applications for risk assessment in other domains. Some critical points in the use of MCS have been stated by Ferson (1995) and Ferson and Ginzburg (1996): Input distributions for example, are usually unknown and therefore uniform distributions of variables are supposed in many cases. This can have considerable effects on the tails of the simulated probability distribution functions (Bukowski et al., 1995). Furthermore, the correlations between different input factors are often unknown, assumed to be static or simply ignored.

2. Selected approaches to consider financial uncertainty

The analysis of decisions under uncertainty has a long tradition. In the eighteenth century it was Daniel Bernoulli and Gabriel Cramer (Bernoulli, 1738) who observed that the expected value of a game was not sufficient to explain the behaviour of the gamblers. They founded the theory of maximization of expected utility, which is today well known as the ‘Bernoulli-principle’. We start our methodological review with the expected utility approach as the broadest concept and continue with closely related concepts.

2.1. The expected utility framework

The Bernoulli-principle proposes to consider the expected utility (EU) for decision making under uncertainty instead of the expected outcome. Therefore, every possible outcome $x_{d,s}$ (dependent on the decision $d \in D$ and the state of nature $s \in S$) has to be evaluated with its utility $U(x_{d,s})$ which is defined by the utility function $U(x)$. Decision makers' preferences have an expected utility representation if the following conditions hold:

$$x_1 \succ x_2 \iff E[U(x_1)] \geq E[U(x_2)] \quad (1)$$

where $U(x)$ is the utility function of random outcomes x and $E[U(x)]$ is the expected value of utility function $U(x)$. The expected utility of decision d is the sum of the utilities weighted with the probability p_s

$$E[U(x_d)] = \sum_{s=1}^S U(x_{d,s}) p_s \quad (2)$$

if the random variable x is discrete; for continuous random variables the sum has to be replaced by the respective integral.

Von Neumann and Morgenstern (1944) showed that a unique cardinal utility function (up to a positive affine transformation) can be derived on the condition that the following set of axioms for a preference relation on a set of probability distributions is satisfied: completeness, transitivity, independence and continuity (Dubra et al., 2004). Nevertheless, it has been shown by several authors (e.g.

Allais, 1953; Ellsberg, 1961; Aumann, 1962; Kahneman and Tversky, 1979; Machina, 1982; Starmer, 2000) since the 1950s that actual decisions are not always consistent with all these axioms and individuals tend to violate the underlying assumptions systematically (Bell and Farquhar, 1986). For this reason some generalized or extended expected utility approaches have been developed (non-expected utility approaches, e.g. prospect theory, dual expected utility; see Starmer, 2000 for more detailed information), but the von-Neumann-Morgenstern expected utility theorem is still considered one of the most fundamental results of the theory of individual decision making (Dubra et al., 2004). However, there are two main difficulties in practice, the knowledge about or the generation of probabilities of possible outcomes and the accurate consideration of the decision maker's preferences.

Savage (1954)—based on Ramsey (1931) and de Finetti (1937)—integrated the notion of subjective probability into the expected utility theory as a quantitative measure of the decision-maker's degree of belief that a specific event will occur (Fishburn, 1982). Therefore he increased the number of axioms, which include an ordering- and several independence-conditions, a continuity-axiom and a dominance-postulate. A more recent trend to face the problem is the development of robust optimisation techniques for expected utility under ambiguous distributions (e.g. Natarajan et al., 2008). In the framework of dual expected utility theory (Yaari, 1987) distortions on the probability distributions are used to characterize the decision maker's attitudes towards risk (Wang and Young, 1998). On the other hand, the common attitudes towards risk, definitions of appropriate utility functions, the necessary assumptions and their consequences have been studied and discussed quite intensely (e.g. Pratt, 1964; Arrow, 1971; Pratt and Zeckhauser, 1987; Gerber and Pafumi, 1998). In general, different types of utility functions are used to describe the attitude of the decision maker towards risk: linear increasing utility functions for risk neutral decision makers ($U'(x) > 0$; $U''(x) = 0$), convex increasing functions for risk seeking ($U'(x) > 0$; $U''(x) > 0$), concave increasing functions for risk avoiding decision makers ($U'(x) > 0$; $U''(x) < 0$) and combinations of them (e.g. third-order polynomial to account for positive skewness; e.g. Kraus and Litzenberger, 1976). As risk aversion is assumed to be a common behaviour, most studies use concave utility functions that can be expressed by e.g. negative exponential utility. The slope of the utility function characterizes the intensity of risk aversion or affection for risk, which can be determined by the Arrow-Pratt measure (Arrow, 1971; Pratt, 1964). Nevertheless, the necessary assumptions regarding risk preferences must be done carefully and should consider that preferences are not always constant over the entire planning horizon, but rather depending on temporal circumstances (e.g. Post and van Vliet, 2006). Furthermore, the real behaviour of investors is not always rational and full information regarding preferences is not always available. Empirical studies to determine utility functions of relevant decision makers in natural resource industries for subjective expected utility analysis (Savage, 1954) have given rather unconvincing results (Anderson and Hardaker, 2003).

In the following paragraphs we will explain the approaches considered in this review and their relationship to the expected utility framework.

2.1.1. Stochastic dominance

Stochastic dominance (SD) is an approach that considers the entire probability distribution of outcomes (Hadar and Russell, 1969). Hence, it is not dependent on normally distributed outcomes and assumptions relating to utility functions are only weak. As a consequence of the less restrictive assumptions or partial information, the results of the approach are less determinable and often only a partial ranking of advantageous alternatives is possible (e.g. Knoke, 2008). However, at least they result a differentiation of efficient and inef-

ficient sets of investments; if there are two or more efficient alternatives, the investor has to decide subjectively which choice to take.

Levy (1998) explained different methods of SD. First degree stochastic dominance (FSD) is an investment decision rule corresponding to partial information; it only supposes that the investor prefers more money rather than less money, that is, the first derivative of the Utility function U is

$$U' > 0 \quad (3)$$

Subsequently, we assume that the random variable x is continuous with probability density $f(x)$ and cumulative probability function $F(x)$:

$$F(x_i) = \int_{-\infty}^{x_i} f(x) dx \quad (4)$$

If we have the cumulative distributions of two different investment alternatives, represented by their cumulative distributions F and G , respectively, alternative F dominates alternative G by FSD iff:

$$\begin{aligned} G(x) - F(x) &\geq 0 \quad \forall x \\ G(x) - F(x) &> 0 \quad \text{for at least one } x \end{aligned} \quad (5)$$

It is thus possible to compare two investment alternatives by their cumulative distributions of outcomes (e.g. financial return). If we have the cumulative distributions of two different investment alternatives F and G , alternative F dominates alternative G by FSD if the cumulative distribution of outcomes of alternative F is equal to or less than that of alternative G for every possible outcome and if at least one single outcome exists, where the cumulative probability is smaller for alternative F . A differentiation of efficient (undominated) and inefficient (dominated) investments is possible and every decision maker with preference for higher returns ($U' > 0$) would agree to this partial ordering. Inefficient alternatives need not be dominated by all efficient alternatives; one single dominating alternative is sufficient to classify the dominated alternative as inefficient.

SD decision rules are commonly used for the initial screening of alternatives and for a partial ordering based on partial information about the decision maker's preferences. Only in exceptional cases remains just one efficient investment as a result. If different efficient alternatives exist, their cumulative distributions of expected financial returns have to intersect; otherwise one of them would dominate. Yet, those alternatives with intercepting distributions cannot be ranked by FSD. Hence an application of FSD is of limited use in many cases. Thus, it is desirable to seek for further criteria.

Second degree stochastic dominance (SSD) does not only assume that the investor prefers more over less. SSD additionally assumes that he/she is a risk averter. Hence, not only the first derivative of his/her utility function has to be positive ($U' > 0$), but also the second derivative has to be negative for every possible outcome ($U'' < 0$).

$$U' > 0 \text{ and } U'' < 0 \quad (6)$$

It can be proved that an investment F dominates another investment G by SSD iff:

$$\begin{aligned} \int_{-\infty}^x [G(x) - F(x)] dx &\geq 0 \quad \forall x \\ \int_{-\infty}^x [G(x) - F(x)] dx &> 0 \quad \text{for at least one } x \end{aligned} \quad (7)$$

SSD requires that the integral of the cumulative distribution function of investment F is equal to or lower for at least one possible outcome x than the corresponding integral of alternative G .

Criteria have also been developed for a third degree stochastic dominance (TSD; Whitmore, 1970) or even higher orders; TSD

assumes in addition to the assumptions of SSD that the third derivative of the utility function is positive, so we have $U' > 0$, $U'' < 0$, $U''' > 0$. It can be shown that these assumptions capture also a preference for positive skewness of probability distributions (a distribution with rare but high outcomes). In this case, the decision maker becomes decreasingly averse to risk as wealth increases (and vice versa) and empirical evidence suggests this attitude for many investors (Post and van Vliet, 2006). However, as previously mentioned, applications of SD can offer only a differentiation of efficient and inefficient sets of investment, but there may only be few inefficient investments. To increase the distinctive power, even higher degrees of SD have been proposed. Unfortunately, the increasingly more restrictive assumptions with respect to the utility function are increasingly hard to justify on economic reasons and the incremental gains are often low. Therefore, scientific studies in decision analysis mostly use SSD (e.g. Kuosmanen, 2004) and applications of TSD in forest science studies are rare. Caulfield (1988) used FSD, SSD and TSD analysis in order to determine the economic rotation of a forest stand, but TSD analysis did not succeed to remove any distribution from the SSD-efficient set. Hence, we do not provide a more detailed explanation of TSD.

Nevertheless, as shown by Post and van Vliet (2006), TSD utility functions also exhibit a strong aversion against downside risk. The following approach considers explicitly the left tails of the distribution functions in order to minimize the risk of falling below a critical outcome.

2.1.2. Downside risk

Most econometric models usually quantify risk as the standard deviation or variance of outcomes. Hence, risk increases if single outcomes differ from the expected outcome, this without considering that positive deviations usually are not recognized as a disadvantage, rather, as an opportunity. In the case of forestry, some studies have already investigated the opportunities of these positive deviations to increase the financial return in connection with flexible timber harvest strategies (e.g. Brazee and Mendelsohn, 1988; Knoke and Peter, 2002). In contrast to the conventional risk measure, downside risk models define risk as expected outcomes below a certain minimum. Hence, these non-symmetric risk measures are based on negative deviations.

So called *lower partial moments* (LPM), which were among other authors investigated by Bawa (1975), Fishburn (1977) and Harlow (1991), is an approach to investigate the risk of falling below a selected critical outcome x^* . It might be called an application of the Safety-First rule for investment decision making. For calculation of LPM, Lee and Rao (1988) used the following equation:

$$LPM_n(x^*; f) = \int_{-\infty}^{x^*} (x^* - x)^n f(x) dx \quad (8)$$

where:

x^* critical reward
 n order of LPM

LPM is calculated as the integral of the weighted return distribution for all financial returns below a critical reward x^* ; the order n defines the weight or whether and how strong negative deviations from the critical reward are to be assessed: If the order n of LPM is set to 0, LPM_0 gives the probability of obtaining an outcome less or equal to x^* (shortfall probability) without considering the amount of shortfall. LPM_1 gives the mean of the shortfalls below the critical reward (shortfall expected value), with equally weighted negative deviations. LPM_2 gives the mean squared deviation from x^* (shortfall variance), with more emphasis on lower values.

LPM can also be used as a risk measure in a Mean-LPM framework (Bawa and Lindenberg, 1977; Fishburn, 1977), where the mean of outcomes is used as a return measure. As shown by Bawa (1975, 1978), portfolio optimisation based on the Mean-LPM model produces (for a generalised location and scale family of distributions) solutions that dominate all other portfolios according to the concept of SD (Grootveld and Hallerbach, 1999). For $n=0$, Mean-LPM₀ optimisation produces first order stochastically dominant results; Mean-LPM₁ optimisation yields second order stochastically dominant portfolios and Mean-LPM₂ optimisation results in third order stochastically dominant portfolios. Hence, the order n affects the utility function consistent with the applied risk measure. LPM₁ is consistent with risk averse utility functions ($U' > 0$; $U'' < 0$) and LPM₂ is valid for risk averse utility functions displaying skewness preferences ($U' > 0$; $U'' < 0$; $U''' > 0$).

Another well known downside risk measure is the Value at Risk (VaR), a quantile-based risk measure that gives the worst portfolio loss that will be exceeded only with some given small probability (typically 1% or 5%). Obviously, both downside risk measures are directly related via the cumulative distribution function of outcomes. For a defined LPM₀, the corresponding VaR can be calculated and vice versa.

As already mentioned, other econometric approaches do not consider risk as the outcomes below a minimum financial return. The most common approaches for financial analysis as well as in forest economics have a symmetric point of view in risk. That is, uncertainty is considered as risk by using the variance of outcomes as a risk measure.

2.1.3. Mean-Variance analysis

The scientific work of Harry Markowitz (1952) is a prominent example for an application of the MV rule⁴ to show financial effects of diversification. Following the mathematical conversion of the proverb “Don’t put all your eggs in one basket”, Markowitz used statistical rules to compute the dispersion of financial returns of mixed investments. Later, his theory was developed by Sharpe (1964) into Capital Asset Pricing Model (CAPM), which is also based on MV optimisation. Although Markowitz’ model was criticized several times due to its restrictive assumptions, it still plays an important role as a theory of economic valuation, and indeed, other modern models are based on this approach.

The basic idea is to compare possible investment mixtures or portfolios by their expected financial returns⁵ and their standard deviations. Summing the proportional expected single returns r_i results in the expected financial return R_p of a portfolio with two or more alternative investments (linear relationship).

$$R_p = \sum_{i=1}^n f_i r_i \quad (9)$$

where:

R_p expected financial return of portfolio
 f_i portfolio weight of a single asset, that is, the proportion of value allocated to asset i divided by the total value of the portfolio
 r_i expected financial return (outcome) of a single asset i
 n number of assets

⁴ Mean-Variance rule: If the expected return on two assets is the same but the second has a higher variance (or standard deviation), then the first will be preferred. Also, if the variance on the two assets is the same but the second has a higher expected return, the second will be preferred.

⁵ Studies in forest economics usually focus on simple returns (instead of log-returns) as they do not investigate time series of financial returns. Therefore we consider simple returns when discussing the distributional properties.

Note that complications may arise if possible interactions between mixed assets have to be considered, which may occur, for example, in biological systems. These interactions can have impacts on both, the mean as well as the variance and therefore they can only partially be considered by correlations. However, these effects are a topic for bioeconomic modelling (see Knoke and Seifert, 2008) rather than for the valuation of financial effects.

The standard deviation of financial returns can then be used to quantify the risk of a mixed investment:

$$\sigma_p = \sqrt{\sum_{i \in N} f_i^2 \sigma_i^2 + \sum_{i \in N} \sum_{j \in N} f_i f_j \text{cov}_{i,j}} \quad (10)$$

$$\sum_{i \in N} f_i = 1; \text{cov}_{i,j} = k_{i,j} \sigma_i \sigma_j;$$

where:

σ_p	standard deviation of portfolio returns (set of risky assets)
i, j	indices for specific assets
N	number of available assets
f_i	portfolio weight of a specific asset
σ_i	standard deviation of returns for asset i
$k_{i,j}$	coefficient of correlation between the returns asset i and asset j
$\text{cov}_{i,j}$	covariance between the returns asset i and asset j

Effects of diversification (depreciation of risk for a given expected return or increase of return for a given risk) can be identified for different combinations of investments, provided that the variability of their financial return is not perfectly positive correlated ($k \neq 1$).

The decision regarding the optimum fraction of the different investments finally depends on the risk tolerance of the investor, expressed by his individual utility function. Note that there might be different understandings of the term “risk aversion” in the context of SD as compared to MV. In the context of SD, risk aversion is interpreted in the classic sense of von Neumann–Morgenstern expected utility and results from a negative second derivative of this utility function. In general, MV is not compatible with expected utility. This is only the case if certain restrictions on the utility function (see e.g. Levy and Markowitz, 1979; Bell, 1995) or on the underlying probability distributions hold (see e.g. Chamberlain, 1983). Most notably, MV is compatible with expected utility if the underlying probability distribution is the normal distribution. In this particular case, the SSD criterion is equivalent to the MV criterion: an investment is preferred if it has a higher expected return and a return variance not higher than an alternative investment or a lower return variance and an expected return not less than an alternative. In those cases where MV is not compatible with expected utility, which includes most cases in practice, risk aversion is interpreted loosely as variance aversion. Nevertheless, even in those cases where MV is not strictly compatible with expected utility, it might allow for fairly decent approximations, which may justify its widespread use in practical applications.

The only form of risk tolerance where diversification effects have importance for the investor is risk aversion. Investors with risk neutral behaviour, for example, will choose the asset with the highest expected financial return without regarding the risk (Spremann and Kotz, 1981). The higher the risk aversion, the greater the utility gain implied by diversification (Wenner, 2002). Hanoch and Levy (1969) state that a portfolio can only be efficient if, and only if an increasing and concave quadratic utility function exists, that rationalizes the selection of the portfolio. A common approximation used to maximize the utility of a risk averting person is to calculate the estimated

certainty equivalent (CE), which is based on a negative exponential utility function (e.g. Gerber and Pafumi, 1998).

$$CE = E(NPV) - \frac{\alpha}{2} * \sigma_{NPV}^2 \quad (11)$$

where

CE	Certainty equivalent
$E(NPV)$	expected NPV
α	Constant quantifying the absolute degree of risk-aversion.
σ_{NPV}^2	variance of NPVs as a measure for uncertainty

The CE represents the value that makes the risk-averse decision maker indifferent between receiving the CE or the corresponding risky prospect. Therefore, the expected financial return (in terms of NPV) is reduced by the risk premium, which is the proportionally weighted variance of NPVs. The weight depends on the absolute degree of risk aversion α which can be estimated by $\alpha = \frac{\gamma}{I}$, with γ representing the relative risk aversion (e.g. $\gamma = 1$ for moderate risk aversion) and I as the initial investment amount (Spremann, 2010). Moreover there are many further ways to calculate the CE for different classes of utility functions (e.g. Gerber and Pafumi, 1998) and to determine the absolute degree of risk aversion (e.g. using the Arrow–Pratt index of absolute risk aversion; e.g. Featherstone and Moss, 1990; Holt and Laury, 2002).

This portfolio selection approach still has importance for financial valuation; however, there has been some fundamental critique on this classical concept and its restrictive and sometimes unrealistic assumptions. Perhaps the most difficult and important assumption is the investor's utility function (see also Section 2.1).

Another problem is the assumption of normally distributed financial returns (e.g. Gotoh and Konno, 2000; Mandelbrot and Hudson, 2005), which, especially in case of forestry, is often unrealistic. Knoke and Wurm (2006) for example demonstrated that for central European stands, there is a considerable deviation from normally distributed simple returns when considering the effects of natural hazards with Monte-Carlo simulations. Bawa (1975) developed mean-LPM portfolio selection rules, which allow for arbitrary probability distributions and use the LPM as a risk measure. As a special case, they also contain the MV rule.

The distribution function of outcomes is also important for the next models. In contrast to the above mentioned approaches these models in general are not related to the expected utility concept but rest on arbitrage arguments.

2.2. Option pricing models

Classic investment analysis has been criticized for not taking into account the activities of decision makers after the investment is put into practice (Kruschwitz, 2009). In fact, there is of course usually the possibility to change at least some details, if and when things go wrong. The amount of flexibility also depends on the initial decision, which can cause different future action alternatives. Therefore, it can be useful to identify potential sources of flexibility within a project. The real options approach is a common way to quantify the economic value of real investment flexibility. As it is based on the theory of financial options, we first want to introduce briefly the notion of the financial options approach.

Financial options are contracts, which include the right to sell (put options) or buy (call options) an underlying asset at a certain price and in a certain period (American options) or at a fixed date (European options). In order to determine the value of an option it is necessary to estimate the possible future development of the underlying asset. The most common models use different random processes for this purpose: the time-discrete binomial model (Cox et al., 1979) is based on

a binomial process, whereas the time-continuous *Black–Scholes-model* (Black and Scholes, 1973) is based on a Geometric Brownian motion (GBM)⁶ process. The *binominal model* value converges to the corresponding *Black–Scholes-model* value for an increasing number of time steps (Cox et al., 1979).

The calculation of the value of a financial option in both models is based on the assumption of the possibility to create a hedged position of the underlying asset and options, which is riskless (replicating-portfolio assumption). Usual model assumptions are that it is possible to borrow or lend money at a constant riskless interest rate, that there are no transaction costs to be considered, that the shares are divisible and that arbitrage opportunities do not exist (assumption of a perfect market in capital). Arbitrage is the possibility of a risk-free profit without costs, in particular, to play on a price difference between two markets by simultaneous transactions. The no-arbitrage assumption has some simple but strong implications for the option value, e.g. that neither the drift parameter of the price processes of the aforementioned models nor the investor preferences enter the pricing equation for the option.

On these conditions, it is possible to calculate the European call option value C in the framework of the *binominal model* in terms of the current stock price S , the exercise price K , the up and down factors u and d for price movements per period ($u \geq 1$; $0 < d \leq 1$) and r as one plus the riskless interest rate over one period of the binomial tree. To avoid arbitrage, ($u > r > d$) must hold. To simplify matters, the basic model ignores possible cash dividends, transaction costs, margin requirements and taxes (Cox et al., 1979).

$$C = S\Phi[a; n; p'] - Kr^{-n}\Phi[a; n; p] \quad (12)$$

where

$$p \equiv \frac{(r-d)}{(u-d)} \quad \text{and} \quad p' \equiv u * p / r$$

$a \equiv$ the smallest non – negative integer greater than $\log\left(\frac{K}{Sd^n}\right) / \log\left(\frac{u}{d}\right)$

$\Phi[\cdot]$ complementary binominal distribution function

The up and down factors u and d can be used to derive transformed, so called risk-neutral or pseudo-probabilities p (p'). With respect to these pseudo-probabilities, the valuation equation is equivalent to the present value of the expected future cash flow of the option. Both factors are usually estimated in terms of the underlying volatility σ and the time T to expiration of the option (Cox et al., 1979):

$$u = e^{\sigma\sqrt{T}} \quad \text{and} \quad d = e^{-\sigma\sqrt{T}} \quad (13)$$

The *Black–Scholes-model* assumes a lognormal distribution of future payoffs and uses a partial differential equation (PDE) to determine the option value. The PDE can be solved to the continuous-time Black–Scholes-formula for a call option (without dividends), which can be used for e.g. forest valuation (e.g. Hughes, 2000; Gjolberg and Guttormsen, 2002).

$$C = SN(x) - Ke^{-rT}N\left(x - \sigma\sqrt{T}\right) \quad (14)$$

⁶ GBM is used to describe the random walk of variables with constant drift and volatility. Alternatively, mean-reverting processes (also known as Ornstein–Uhlenbeck processes) with prices fluctuating stochastically around a long-run mean have been used (Amram and Kulatilaka, 2003, p. 93; Hull, 2009, p.482).

where

$N(\cdot)$ standard normal distribution function

$$x = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

r the continuously compounded risk-free rate

Due to the no-arbitrage assumption, there is a direct relationship (*put-call parity*) between the European call option value C and the European put option value P (Merton, 1973).

$$P + S = C + Ke^{-rT}$$

The put option value P plus the current stock price S must be equal to the call option value C plus the present value of exercise price K and vice versa. Otherwise there would be arbitrage opportunities.

Further extensions allow for less restrictive assumptions, e.g. taking into account constant dividend payments (Cox et al., 1979; Merton, 1973) or allowing for sudden fluctuations of stock prices as Poisson driven jumps (Merton, 1976). Bates (2003) provided a review of the developments in empirical option pricing research and alternative models. He stated a substantial focus on stock index options and the difference of implied volatilities across various strikes (volatility skew). Finally, he suggested a renewed focus on the financial intermediation of risk and encouraged further studies on equilibrium option pricing models that incorporate model and parameter uncertainty.

The real options approach can be interpreted narrowly as an extension of financial option pricing models to the valuation of non-financial assets (Amram and Kulatilaka, 2000). Myers (1977) introduced the term *real options* as an application of option pricing theory to the valuation of non-financial or real investments. Later he classified this approach as a link between strategic planning and financial theory but he also recognized the complexity of real investments and the arising difficulties when applying financial theories for valuations (Myers, 1984). According to Trigeorgis (2004) a classification of different kinds of common real options can be done: options to defer, time-to-build options, options to alter the operating scale, options to abandon, options to switch, growth options and multiple interacting options; however, for every category of the real options approach, when applied to the analysis of physical assets there are a couple of differences to the analysis of financial options which have to be taken into account, e.g. the longer maturity or the fact that the competition, the market and the management assumptions and actions drive the strategic option value (Mun, 2006, pp. 109–112). Therefore, some further extensions of real options approaches in terms of analytical and numerical approaches were developed (Baecker et al., 2003), which have more or less practical relevance and are not discussed in detail in this study. Actually, the commonly used models with high practical relevance are still the *binominal model* (in terms of Decision-Tree-Analysis)⁷ and the *Black–Scholes-model* (Amram and Kulatilaka, 2003) in general, for the valuation of real options as well as in forestry related studies in particular.

Option pricing models can provide a different and useful perspective on uncertainty: They help to clarify the economic value of flexibility in our decision making processes. Uncertainty resolves as time passes and as new information becomes available, options provided by flexibility may be exercised to the advantage of the holder of these so-called real options. The term *real options* refers to the lines of thought which have been developed to value financial options. To view flexibility within real investments from the perspective of option

⁷ Trigeorgis (2004) provides more detailed information.

theory is very useful from a conceptual point of view as it certainly helps to identify potential sources of value within a real project, which are difficult to capture with simpler approaches like net present value calculations. Another issue is the actual valuation of such a real option. In many potential applications it is questionable if the arbitrage pricing arguments which have led to very successful valuations of financial options can be applied. Arbitrage pricing arguments are based on more or less perfect markets, at least with respect to those assets, that are part of the portfolios in partial arbitrage equilibrium. It is an empirical question if these assumptions are fulfilled in a certain application. It might well happen that tradability is limited or that the data necessary to estimate the required model parameters is insufficient (Borison, 2005; Götze et al., 2008). Even if such considerations lead to the conclusion that arbitrage techniques are not applicable, the viewpoint of option pricing might still be advantageous if the resulting model is approached with dynamic programming techniques. Basically, the real options model can increase the attention on flexibility of investments and provide a valuable insight for investment analysis in the context of balancing intuition and reflection (Baecker et al., 2003).

The model interprets flexibility as a range of possible corporate activity. Flexibility as a systems ability to deal with unexpected developments is an important factor for the next concept. It seeks robust solutions providing adequate outcomes even for the case of uncertain data in connection with unpredicted changes.

2.3. Robust optimisation

Robust optimisation techniques have recently been developed and can be used in various areas (e.g. portfolio optimisation, inventory management, mechanical engineering, etc.) in order to consider optimization problems for which it is only known that the data belongs to a given uncertainty set, but the optimization problem's constraints mustn't be violated for all possible values of the data contained in the uncertainty set (e.g. Ben-Tal and Nemirovski, 1998; Ben-Tal et al., 2009). These techniques are thus based on a description of parameter uncertainty via sets that are used in lieu of probability distributions (Ben-Tal et al., 2006b) so as to ensure required outcomes for possible realisations of the true parameters for given bounded uncertainty sets (Ben-Tal et al., 2006a). Soyster (1973) used this concept first in a theoretical framework, but the application lacked attractiveness due to the overly conservative nature of the provided solutions. Mulvey et al. (1995) integrated goal programming formulations with scenarios of input data in order to generate less sensitive solutions. Further less conservative approaches have been developed by e.g. El-Ghaoui et al. (1998), Ben-Tal and Nemirovski (2000) and Bertsimas and Sim (2004). Ben-Tal et al. (2006a) developed a technique that also allows for data deviation outside the uncertainty set for a controlled loss in performance. Beyer and Sendhoff (2007) provide a recent survey of robust optimisation approaches.

We will discuss *information-gap decision theory* (IGDT; Ben-Haim, 2006) in greater detail, because this technique has obtained the greatest popularity in ecosystem management so far. However, we also acknowledge the criticism to IGDT (Sniedovich, 2007), which classifies IGDT as a special instance of Wald's maximin model and as a local sensitivity analysis (see Section 4). Information-gap models investigate the lack of information, or the gap, between what is known and what needs to be known for the optimal solution. Decision makers can be assisted in developing preferences, assessing risks and opportunities, or choosing sources of information and lines of exploration, in light of the analysis of severe lack of information (Ben-Haim, 2006). From the perspective of IGDT, a "robust" model-based decision is characterized by great immunity against adverse uncertainty that may affect the parameter estimates. I.e., the sensitivity of the decision results to changes (errors) in the estimated

parameters is low with regards to the achievement of a minimum required outcome (performance criterion). That means a "robust" decision achieves the predefined performance criterion over a possibly large uncertainty space. The Info-Gap robustness models first and foremost aim at maximizing the immunity of model-based economic decisions to adverse uncertainty (Ben-Haim, 2010). They thus strive to maximize that specific size of a possible uncertainty set, which would still allow for acceptable solution outcomes.

Information-gap models can be realized in many different forms. This is due to fact that uncertainty can have different facets and the lack of information may lead to unexpected losses as well as gains and the decision maker's preferences and ambitions may differ over time so that the value of information does not remain constant. Ben-Haim (2006) describes different aspects of IGDT and their relation to classical decision models. Proposed model shapes include simple boxes or confidence ellipsoids. As in the MV model, the lack of information might offer propitious opportunities as well as pernicious dangers of failure. In contrast to the classical approach, some information-gap models work with non-probabilistic quantification of uncertainty (Ben-Haim, 2006). Clusters of events representing different levels of information and their effects on the decision are investigated.

A financial information-gap model is a cluster of sets of possible variation around the momentary knowledge $u^{(t)}$ (e.g. a calculated probability distribution, based on historical data) and with the unknown range of an uncertainty parameter α , which characterizes the size of the uncertainty set. Therefore the cluster of nested sets can be formulated as:

$$U(\alpha, \tilde{u}) = \{u(t) : |u(t) - \tilde{u}(t)| \leq \alpha\}; \alpha \geq 0 \quad (15)$$

This equation is quite general and Ben-Haim (2006 p. 20–30) derived some more specific information-gap models in which the structures depend on the available prior information.

One basic function of IGDT is the assessment of a possible uncertainty set that is specific for every chosen decision vector (or vector of control variables), f , which would still allow for achieving the minimum required reward, y^* . In order to make decisions that minimize the risks to experience an undesired outcome, the robustness function assesses the immunity to failure below a minimum required reward.

$$\hat{\alpha}(f, y^*) = \max\{\alpha : (\min_{u \in U(\alpha, \tilde{u})} R(f, u)) \geq y^*\} \quad (16)$$

where:

$\hat{\alpha}(f, y^*)$	robustness
f	decision vector
y^*	minimum required reward (performance criterion)
α	uncertainty parameter
$R(f, u)$	reward function

The robustness function seeks the maximization of immunity to uncertainty, or, the highest level of an uncertainty parameter α at which minimal requirements are always satisfied and failure cannot occur. Thus, $\hat{\alpha}(f, y^*)$ is the greatest level of uncertainty that still guarantees not less than the critical reward y^* . The reward function $R(f, u)$ represents the success of the decision, which depends on the decision vector f as well as on the uncertainty vector u , whose variation is characterized by the cluster of sets of possible variation $U(\alpha, \tilde{u})$, with $\alpha > 0$. The minimal reward y^* should be no less than the critical reward with maximal uncertainty level, such that the minimal requirements are satisfied in the most unexpected case. A decision alternative with a higher value of robustness at the same level of critical reward is therefore preferred.

There are many different opportunities for applications of this theory. Often, the robustness of an estimated probability density function (PDF) should be evaluated by confrontation with unknown uncertainty. The danger of uncertainty is an undesirable financial outcome (financial failure), and for this reason a minimal critical reward needs to be defined. Considering the extremely long production periods of forestry in connection with future uncertainty, the question is, how much confidence do we have in the estimated probability distribution, or, how wrong can the estimated PDF be so as to still offer a minimal critical reward at a specific level of statistical confidence?

According to Ben-Haim (2005), who evaluated the robustness of an estimated financial Value at Risk (VaR) for some selected statistical confidence levels under the assumption of a standard normal distribution, the robustness of a PDF can be derived from the following principle equation:

$$\hat{\alpha}(y^*, P_c) = \max \left\{ \alpha : \left(\max_{p \in U(\alpha, \hat{p})} P_f(p) \right) \leq P_c \right\} \quad (17)$$

Here, the robustness function describes the highest level of an uncertainty parameter α at which the maximum probability of failure $P_f(p)$ is still not greater than the critical (accepted) probability of failure P_c . Thus, $\hat{\alpha}(y^*, P_c)$ is the robustness to uncertainty in the estimated PDF of probabilistic information. It may be computed as follows:

$$\begin{aligned} \hat{\alpha}(y^*, P_c) &= \frac{P_c}{P_f(p)} - 1 & \text{if } P_f(p) \leq P_c \\ \hat{\alpha}(y^*, P_c) &= 0 & \text{if } P_f(p) > P_c \end{aligned} \quad (18)$$

where:

$\hat{\alpha}(y^*, P_c)$ estimated robustness depending on minimum acceptable financial return y^* and the accepted probability of failure P_c
 $P_f(p)$ estimated probability for a financial return r below the minimum financial return y^* : Probability of failure
 P_c tolerated error, with $1 - P_c$ quantifying the statistical confidence

It is clear that demanding a small failure probability P_c automatically means low robustness and that the robustness is zero if the estimated probability of failure $P_f(p)$ exceeds P_c . We may thus say that a high required performance will always trade off against robustness.

If no other information is available one can assume normally distributed financial returns (e.g. Ben-Haim, 2005) and estimate $P_f(p)$ as follows:

$$P_f(p) = \Phi \left(\frac{y^* - R}{\sigma_r} \right) \quad (19)$$

Ben-Haim (2006) also presents an example of an application for portfolio investment. The robustness $\hat{\alpha}(f, y^*)$ of the portfolio investment can be calculated as (formula with alterations):

$$\hat{\alpha}(f, y^*) = \frac{f^T \hat{u} - y^*}{\sqrt{f^T W f}} \quad (20)$$

where:

$\hat{\alpha}(f, y^*)$ estimated robustness depending on decision vector f and the minimum acceptable financial return y^*
 f^T transposed (line) vector of asset fractions (decision vector)
 f column vector of asset fractions
 $\hat{u}$ nominal financial return vector
 W covariance matrix

The robustness is defined for non-negative results, otherwise the robustness is zero.

If no other information is available, the term $f^T \hat{u}$ has to be calculated as the mean of all possible returns and the term $\sqrt{f^T W f}$ as the standard deviation of all possible returns, while considering the variance–covariance structure of the portfolio. The ratio introduced by Ben-Haim (2006) is structurally analogous to the reward-to-variability ratio first derived by Sharpe (1966).

There is no link between optimisation in the framework of IGDT and EU-theory, but the EU-model is included in IGDT as a special case (Ben-Haim, 2006, pp. 193–205). IGDT models can be used to incorporate an uncertainty horizon about probabilities and utilities in the EU-framework (e.g. Regan et al., 2005). Furthermore, several studies have developed robust optimisation techniques for the classical MV portfolio allocation problem (e.g. Goldfarb and Iyengar, 2003; Fabozzi et al., 2007), which can be linked to EU-theory. In this context Fabozzi et al. (2007) state that the size of the uncertainty set “controls the aversion to the uncertainty that comes from the estimation error”.

Applications of robust optimisation in forestry or other land use applications are still rare. Yet, there exist some applications in biological conservation. Some applications of the mentioned approaches in forestry will be presented in the following overview.

3. Forestry studies and financial uncertainty

Investigations concerning economic asset valuation under uncertainty within the scope of forestry are still rather scarce even though research interest has been increasing in the last decades. Some investigations deal with this question in relation to the assessment of agroforestry systems and non-timber forest products (e.g. Blandon, 1985; Lilieholm and Reeves, 1991; Babu and Rajasekaran, 1991). In the general field of ecosystem investigations Figge (2004) noticed that although the correlations between diversity and stability have been discussed several times, the parallels between approaches of financial mathematics and ecology have generally not been acknowledged. In his study he presents the fundamentals of portfolio theory with an application to biodiversity. He explains the link between diversification and diversity: The elements of a portfolio (or an ecosystem) provide returns (or benefits which society derives from a species) which are subject to uncertainty and portfolio theory can be usefully applied to assess biodiversity. The following paragraph provides summaries about this link between diversification and diversity in forestry literature and about further studies which apply the previously explained approaches for financial assessment under uncertainty.

3.1. The expected utility framework

3.1.1. Stochastic dominance

Various general academic investigations as well as considerable empirical applications of SD have been done up to this point, although in the context of forestry or biodiversity, there exist only a few studies, which use SD. Caulfield (1988) used FSD and SSD to determine the optimum economic rotation of a forest stand for risk-averse decision makers. He considered risk of fire occurrence, compared the results of SD- to those of MV-optimisation and suggested that SD analysis is more useful as a screening technique for the rotation decision under risk. Benitez et al. (2006) investigated appropriate compensation payments for biodiversity conservation using the SD approach to take into consideration the aspect of uncertainty of land use revenues. They compared empirical distribution functions of returns from a risky but ecologically preferred agro-forest land use (shaded coffee areas) with those from agricultural alternatives and calculate the required compensation payments. The results for required compensation under risk are clearly different from the calculated mean values (risk neutral

assumption), which stresses the importance of considering uncertainty in this case. Knoke et al. (2008) used FSD and SSD for derivation of adequate compensation payments for the conservation of a secondary forest. They calculated an effective amount of compensation, which is convincing for almost every decision maker but considerably lower than the necessary amount based on FSD. Further applications of SD in the context of forestry were made by Heikkinen and Kuosmanen (2002), who investigated the effect of considering possible timberland trades on the optimal harvesting problem of a forest holding, as modelled by Heikkinen (1999) with MV analysis. Using empirical data from Finland, Heikkinen and Kuosmanen (2002) assessed the usefulness of two alternative linear programming test approaches (see Kuosmanen, 2001, 2004; Post, 2001, 2003). They analysed the efficiency of portfolio combinations consisting of four harvestable mixed forest stands and the average index of finish stock exchange (Helsinki Stock Exchange, HEX). In the theoretical case of possible purchases, the efficiency changed quite intensively and an 11 percentage points higher mean return became possible. Finally, Heikkinen and Kuosmanen (2002) confirmed the straightforwardness of both alternative approaches and proposed further application of SD, with consideration of catastrophic events and sensitivity to possible sampling errors.

Knoke (2008) used SSD analysis for financial optimisation of tree species portfolios and compared their results with MV analysis and the IGDT approach. In case of a moderate relative risk aversion MV analysis favoured a balanced mixture, meanwhile SSD tended to a higher proportion of the species with lower expected NPV and lower uncertainty but was only able to distinguish between efficient and inefficient mixtures. Nevertheless, the independency of SSD from assumptions about outcome distributions turned out to be advantageous.

Lien et al. (2007) used a related approach (stochastic efficiency analysis with respect to a function; Hardaker et al., 2004) to analyse an optimal tree replanting strategy for a recently harvested stand under stochastic timber prices and volume growth. The method allows for simultaneous comparison of alternatives in terms of certainty equivalents for a specific range of risk attitudes. Therefore the efficient set can be considerably reduced.

In conclusion, SSD could be seen as a method to exclude inefficient alternatives, while a ranking among efficient alternatives must be carried out with other approaches. This restriction is related to the advantageous properties of using general conditions on utility functions and accounting for the entire probability distributions without restrictions on their form. Therefore, natural resource management is a promising application area for SD as deviations from normal return distributions are often observed (Kuosmanen, 2004). Nevertheless, the experiences with SD approaches in forest economics are still low.

3.1.2. Downside risk

Although downside risk models seem to be particularly suitable for financial assessment in forestry, the applications are also scarce in this case. Ramirez et al. (2001) investigated returns, stability and risk of agroforestry systems and monocultures in order to evaluate the effects of diversification of agroecosystems. They defined risk as the probability of falling below a required minimum family income in the area of their investigation and stability as the variance of NPVs. Agroforestry systems showed higher expected NPVs and lower risk levels than the corresponding monocultures, even if the variance as stability measure of the agroforestry returns were larger. Another example is the simulation of frequencies of negative net present values (NPV) by Knoke and Wurm (2006). As already mentioned they simulated NPV frequencies for stochastic timber prices and hazard events for different mixtures of spruce and beech and compared the simulated distributions with expected (normally distributed) NPV frequencies. Simulated frequencies for negative NPV were lower than

the expected for stands with higher proportions of beech, which leads to an underestimation of the optimum proportion of beech when applying the MV rule. For both cases of simulated and expected distributions, mixed stands minimized the probability of negative NPV. Beinhofer (2010) investigated the effects of diversification for different assortments (saw timber and pulpwood) in Norway spruce and Scots pine stands using the MV approach (certainty equivalent) as well as the quantile-based VaR measure. He indicated that producing different assortments of spruce and pine leads to risk compensation, especially when considering portfolios consisting of optimised stands. Optimisation according to downside risk resulted in more conservative portfolios due to a higher risk aversion when using the VaR approach.

Downside risk models provide useful valuation techniques with a special focus on safety. Therefore, they seem especially suitable for valuation in forestry and ecosystem management with regard to sustainability assessment and the included precautionary principle.

3.1.3. Mean-Variance analysis

The majority of the existing studies regarding financial valuation in forest science deal with approaches of mean-variance optimisation, such as portfolio selection approach or its extension to the capital asset pricing model (CAPM). Penttinen and Lausti (2004) give a short review of relevant approaches in forest science literature existing up until 2004. These studies mainly discuss topics like the effects of risk aversion on the attractiveness of forest investments or harvesting decisions (Olsen and Terpstra, 1981; Redmont and Cubbage, 1988; Lausti and Penttinen, 1998; Heikkinen, 1999, 2002, 2003; Heikkinen and Kanto, 2000; Gong and Löfgren, 2003) or the effects of the integration of woodland as an asset in a mixed portfolio investment (Mills and Hoover, 1982; Wagner and Rideout, 1991, 1992; Zinkhan et al., 1992; Thomson, 1997; Duffner, 1999; Lönnstedt and Svensson, 2000; Wippermann and Möhring, 2001; Zinkhan and Cubbage, 2003; Penttinen and Lausti, 2004).

In these studies, forest investments seem to be an attractive alternative for risk diversification and the integration of forest investments in portfolios mostly appears advantageous. Lönnstedt and Svensson (2000) for example have proven negative correlations between forest holdings and other investment alternatives for non-industrial private forest owners in Sweden. The MV approach has been applied successfully piece by piece. For instance, the one period asset management model by Heikkinen (1999) was extended to a multi-period model with stochastic optimisation by Heikkinen (2003), both considering harvesting instead of land sales (e.g. Mills and Hoover, 1982) as a more realistic scenario.

There has been some investigation into diversification effects in forestry production. Reeves and Haight (2000) calculated optimal shares of sawtimber and pulpwood production for loblolly pine stands in the South East of the USA. Provided that 55% of the area is used to produce short rotation sawtimber and 45% of the area is used to produce short rotation pulpwood, their example demonstrates a possible risk reduction of 80% (compared to the production of 100% sawtimber with higher rotation age). In this case the risk reduction is based on smaller revenue variances because of short rotation management as well as low correlation between sawtimber and pulpwood prices. However, the risk reduction measures reduced expected revenues by 50%.

Some approaches of portfolio theory also deal with aspects of mixed forests and converge to financial assessment of biodiversity. Thomson (1991) demonstrated financial advantages for mixed stands in relation to pure stands of the apparently most profitable species. He calculated an optimal mixture, under the assumption of moderate risk aversion, of about 56% southern pine, 27% southern ash and 16% white pine in the South and Midwest of the USA. In Central Europe, Weber (2002) recommended mixtures of Norway spruce and European beech because of the low risk correlation of these species. Knoke et al.

(2005) investigated diversification effects for Central European mixtures of Norway spruce and European beech and calculated a mixture of 70% spruce and 30% beech to obtain maximum utility for a risk-averse investor. When considering timber harvests as dependent on timber prices, the optimal share of beech (the less profitable species in this case) increases to more than 50% (Knoke and Wurm, 2006). Hildebrandt et al. (2010) investigated optimum proportions of mixed species plantations in Chile accounting for the risk of timber price fluctuations. They found relatively low diversification effects but mixed stands proved generally more advantageous than pure stands.

Thomson (1989) evaluated tree improvement investments considering uncertainty by using the CAPM. In the case of West coast Douglas fir, financial risk in relation to the rate of return was quite small and the investment seemed worthwhile.

MV approaches have been well introduced to forest economics for financial valuation under uncertainty. Typical applications concern effects of diversification at various levels (products, tree species or forestry in investment portfolios). Nevertheless, the link between biodiversity and financial diversification for example appears still to require further research.

3.2. Option pricing models

The suitability of option pricing models for real investments have been discussed critically for different reasons (e.g. the usual non-tradability of investment projects, problems in determining a duplication portfolio, see Götze et al., 2008, p. 307; Kruschwitz, 2009, p. 444); nevertheless, some authors of forest science studies (e.g. Zinkhan, 1995; Thorsen, 1999; Duku-Kaakyire and Nanang, 2004) claim that traditional economic approaches ignore the importance of flexibility for the evaluation of forest resource management, leading to erroneous results. There have been some interesting studies using option pricing models to value management options in forestry. In addition, the well-known reservation price approach (e.g. Brazee and Mendelsohn, 1988; Lohmander, 1987) basically is another real option approach to the harvest decision (Thorsen, 1999).

The appropriate price process in the framework of the Black-Scholes-model has been discussed in several studies concerning option pricing in forestry. A detailed review on this discussion is given in Insley and Rollins (2005), who left this question open for ongoing research. Clarke and Reed (1989) and Reed and Clarke (1990) showed that over a single rotation period neither optimal cutting age nor optimal cutting size depend on the absolute level of timber prices for the case that prices follow a GBM and harvesting costs are ignored. As shown by Yin and Newman (1995) and Insley (2002), this independency is no longer valid if harvesting costs are included in the model. Haight and Holmes (1991) and Plantinga (1998) investigated the role of option values for optimal rotation ages by comparison of different price processes. Plantinga (1998) stated that optimal harvest decisions depend strongly on the nature of the considered price process. In order to describe the consequences for optimal rotation and forest valuation, Gjolberg and Guttormsen (2002) also compared the GBM and a mean-reverting price development. They argue that timber prices do not move randomly but follow a mean-reverting development and give a plausible explanation for the common pricing with low discount rates in forestry, based on the included option value with mean-reverting prices. Both price development processes have been compared in further studies for the optimisation of tree harvesting decisions (Insley, 2002) or the assessment of forest concessions (Rocha et al., 2006). In the latter study, the resulting option values based on GBM were considerably higher and the authors recommended applying the more conservative mean-reverting process, as used for other natural resource commodities. Insley (2002) as well considered the mean-reverting process as more realistic for the behaviour of commodity prices.

Malchow-Møller et al. (2004) provided further methodological advancement, developing a two-option approach with adjacency constraints for the optimal harvest rule. Using an approach with incompletely exclusive options of harvesting and stochastic returns enabled for the assessment of uncertain costs that may result from the constraint of being allowed to harvest adjacent stands after a specific time period. The costs of the constraints increased with the imposed time delay and uncertainty. Jacobsen (2007) introduced a sequential two-option approach for the choice of when to harvest a stand and for regeneration species selection. As the first decision depends on the stochastic values of the present and the future stand, a nested problem with two loops of depending decisions was formulated and solved for numerically. Insley and Rollins (2005) specified the optimal harvesting problem in a multirotational framework as a linear complementary problem, which can be solved numerically and provides higher accuracy than the common models used before. Moreover, they assessed the impact of policy constraints on the option value of harvesting in accordance to price developments. Saphores (2003) used option pricing concepts in order to optimise time continuous harvest decisions (partial and total harvests) under stochastic growth and included the risk of extinction.

With regard to contents, most real option approaches in forestry economics have been focused on the optimal harvesting problem and forest valuation: Morck et al. (1989) used the contingent claims approach in order to determine the rotation age for a forestry lease with stochastic growth and timber prices. For a fixed time horizon, they formulated a nonlinear partial differential equation to be solved by numerical methods along a set of constraints and applied the model to a hypothetical white pine forest. Based on the binominal option-pricing model, Thomson (1992) calculated optimal rotation ages with uncertain timber prices and timber yields and assessed the value for the choice of abandonment after harvesting. Finally, he compared the results with the classical Faustmann model. Considering timber price volatility and managerial flexibility, the resulting optimum rotation periods were generally longer (except for high stumpage prices). Stand values were positively influenced by recognition of flexibility especially for volatile and low timber prices. Similar results have been obtained for an example of Japanese stands by Yoshimoto and Shoji (1998) and Yoshimoto (2002) using a combination of stochastic dynamic programming and binominal option pricing with a discretised GBM process; Yin and Newman (1996) analysed the effects on forest investment for timber prices and growth following a GBM and incorporated natural hazards with a Poisson jump process. The catastrophic risk generally decreased the investment value and increased the threshold of investment in forestry projects. Nevertheless, in comparison to the classical Faustmann model, the inclusion of hazard risks has considerably lower effects in a real options model as shown by Insley and Lei (2007). Thorsen (1999) modelled the option value for the establishment of a new forest stand with future timber prices following a GBM and the implications for the appropriate level of afforestation subsidies. In general, taking into account the option value of afforestation opportunities the amount of convincing subsidies increased. Further studies on forest valuation with option-pricing models have been done by e.g. Hughes (2000), Yin (2001) and Yap (2004).

Jacobsen and Thorsen (2003) used option pricing models to develop optimal thinning strategies for mixed-species stands under uncertain tree growth due to possible climate changes. If adaptive management is possible, the mixing of tree species increased the option value even if a species was less productive for climatic reasons. Duku-Kaakyire and Nanang (2004) used the binominal option-pricing model to evaluate an option to delay reforestation, to expand the size of a wood processing plant, to abandon the plant if timber prices fell below a minimum value and a combination of those three options. The options resulted as highly valuable if exercised meanwhile the analysis of land expectation values in terms of a continuous time

version of the Faustmann formula rejected the investments as unprofitable. The authors state that the assessment of forestry projects with static financial analyses is biased for not taking into account price uncertainties and managerial flexibility.

Some studies have also been done specifically on biodiversity valuation (e.g. Weikard, 2003; Kassas and Lasserre, 2004; Leroux et al., 2009), land-use conversion (e.g. Zinkhan, 1991; Reed, 1993; Abildtrup and Strange, 1999; Schatzki, 2003), preservation of an old-growth forest (Conrad, 1997), long-term timber contracts (Shaffer, 1984) and wildlife management (Morgan et al., 2007).

As shown in this section, Option pricing models have been applied in forest economics quite extensively and for a broad range of uncertain and irreversible decisions. They are suitable and close to reality insofar as taking into account the managerial flexibility, but the limitations for the assessment of real investments in forestry have been discussed only rarely.

3.3. Robust optimisation

Approaches based on IGDT have been used in forestry just as rarely. In the case of conservation management, several studies do exist, in which this approach was used to consider uncertainty. Regan et al. (2005) analysed management options for conservation of endangered species by maximizing the robustness of decisions under uncertainty with regard to probabilities and utilities in the EU-framework. All three considered decision-alternatives were affected by uncertainty and their ranking changed with the amount of uncertainty as well as with the considered minimum expected utility. The higher the critical expected utility, the lower the decision's robustness to uncertainty. Regan et al. (2005) stressed this fact as a default in standard decision theory based on maximization of EU and recommended IGDT for a broad range of conservation decision problems. Moilanen and Wintle (2006) investigated the negative effects of fragmentation on the biological value of reserves. An analysis of robustness concerning the occurrence and long running persistence of particular animal species favours aggregated reserves and the application of IGDT proved suitable as a systematic framework for making decisions under severe uncertainty. Furthermore, Moilanen et al. (2006) investigated the consideration of uncertainty with IGDT in general planning of reserve structures. They presented possibilities to grapple with problems in areas with low information certainty brought about by negative (robustness-function) surprises and potentials brought about by positive (opportunity-function) surprises. A comparable study by Halpern et al. (2006) for marine reserve design also highlighted the potential of IGDT to evaluate the costs and benefits of reducing uncertainty by additional data collection.

Deines et al. (2007) analysed the effects of different levels of harvesting on the robustness to uncertainty for a structured long-term population growth model. For the case of an applied population example they demonstrated the interactions between uncertainties of different parameters and the danger of erroneous decisions due to uncertain data. They constitute IGDT as a useful approach for the assessment of management decisions on the risk of extinction, without the need to value each species in the case of differing cost of management decisions. A further study on robust decision-making in conservation management was conducted by McDonald-Madden et al. (2008), who explored optimal strategies for resource allocation with uncertain extinction probabilities and conservation funding.

Apart from conservation management, McCarthy and Lindenmayer (2007) applied IGDT for the assessment of two alternative solutions for replanting the catchment area for urban water supply. Exotic pine plantation and native revegetation were compared with regard to uncertain financial performance of timber production and water yield. The authors showed how the solution most robust to uncertainty depended on the required financial performance as well as on costs of replanting.

Knoke (2008) analysed the robustness of tree species portfolios and stressed IGDT as a useful supplement to classical approaches; moreover he pointed out the advantage of enabling for the quantification of immunity against failure. Results indicated mixed stands as being more recommendable, but beside the assumptions of minimum requirement and accepted probability of failure, the assumptions of the underlying distribution function of outcomes turned out to have high impact on optimum solutions. Further investigations on tree species mixtures performed by Hildebrandt and Knoke (2009) applied a different robust portfolio optimisation, in part obtained from Goldfarb and Iyengar (2003). Hildebrandt and Knoke (2009) considered uncertainties for expected net present values of tree species portfolios as well as their variances and correlations. The tested variability of results for different scenarios showed a lower spread in case of worst-case optimisation including uncertainty bounds. Moreover, the worst-case outcomes of optimum solutions with consideration of uncertainty bounds turned out to be higher for a simulated reference data set. Based on the model of Bertsimas and Sim (2004), robust harvest scheduling decisions for uncertain timber yield and demand have been optimized by Palma and Nelson (2009). They stated a moderate trade-off between robustness and conservatism of solutions and considered the methodology useful for stable decisions in the face of uncertainty. Babonneau et al. (2010) provided further examples for applications of recent developments in robust optimisation in environmental decision-making.

Robust optimisation techniques take into account the uncertainty involved in long-term decision-making so common in forest and ecosystem management. Nevertheless, in general, applications up to now are scarce as the techniques only recently evolved; most have been performed in the IGDT framework, but other developments in robust optimisation seem promising as well for decision making in forestry or ecosystem management. We can be sure they will be applied more frequently and for different purposes in future studies.

To summarize our review of modern approaches used to assess forest or ecosystem investments, we must remark that a lack of investigation still exists in this field. A few approaches of portfolio theory have been carried out, but rather seldom have diversification effects been investigated. Even less frequent are applications of the presented alternative approaches from the expected utility framework (stochastic dominance, downside risk), in general as well as in connection with species mixtures or biodiversity. Given a worldwide trend of an increasing expansion of plantations (Carnus et al., 2006) and comparatively low acceptance of mixtures by investors (Nichols et al., 2006), a more intensive investigation in this area, as well as integration of modern financial approaches is absolutely useful and necessary. Certainly the real options approach has been most frequently used, in the majority of cases to solve the optimal-harvest problem and for forest valuation. Considerable methodological progress has been achieved, even if the complexity of the issues still demands further effort. As robust optimisation techniques are still developing in decision theory, their applications in forestry and ecosystem economics are quite rare. Nevertheless they appear to be promising, especially with regard to severe uncertainty due to extremely long planning horizons, wherein the underlying theory is relatively understandable and communicable. Thus, they will play an important role in general for environmental decision making under uncertainty (Filar and Haurie, 2010).

4. Discussion

Considering the extremely long production and investment periods in forestry, a more intensive reflection of investment risks appears necessary. However, in reality too few scientific studies concerning financial risk quantification of forest investment exist. Just a small portion of them investigate the interrelation between species diversity and risk diversification. In practice, this is

sometimes neglected or almost intuitively executed, as the classic forest economic approach according to Faustmann ignores any appearing uncertainties. Therefore, the principle of precaution as a part of the sustainability term (Endres and Querner, 2000; Knoke and Mosandl, 2004) does not appear to be sufficiently considered. Investigations regarding the effects and value of diversification have been done almost exclusively from an ecological point of view up to now. Drechsler et al. (2005), for example, compared ecological and economic models for biodiversity conservation and stated that the economic models often ignore space, dynamics and uncertainty.

As previously mentioned, most scientific studies that consider uncertainty in regard to forest investment confine themselves to the portfolio selection approach or mean-variance analysis. Their weak points regarding the practical use have been mentioned and alternative assessment techniques have been explicated. These appear to be able to compensate for the weak points of the portfolio selection approach in part, but unfortunately they cannot solve the problems completely, which evolve with the long-term uncertainty of forest investments:

First of all, the generation of a realistic density function of estimated returns is the most difficult task. To avoid sampling errors, large amounts of high quality empirical information regarding price development in the past, as well as the quantification of specific natural risk factors is necessary. However, even with perfect quantification of the past circumstances, the projection to future developments is not necessarily reliable and some further uncertainty remains.

The appearance of unpredictable catastrophic events, technological innovations, or simply changing trends in customer's behaviour, can lead to unexpected changes in price development and correlation between the investment alternatives.

The SD approach, compared with mean-variance optimisation, has some advantages due to the less restrictive assumptions (consideration of the whole distribution of expected returns, independence from the form of the distribution function, weak assumptions regarding the investors utility function), although only a partial ranking of investment alternatives is possible. Therefore, Leshno and Levy (2002) have developed criteria for stochastic dominance, which probably can facilitate a better ranking of the results for most decision makers.

A further difficulty of SD approach is the comparison of portfolios with increasing numbers of choice alternatives due to a relatively high computational burden. Linear Programming tests for SD efficiency (a given portfolio compared to all possible portfolios of a given set of assets) seem to be able to remove this limitation (e.g. Post, 2003; Kuosmanen, 2004; Post and Levy, 2005). Another possible solution with multivariate tests for SSD and TSD efficiency was developed by Post and Versijp (2007). Finally, there remains quite a strong dependency on the distribution of financial returns. Small samples can cause large sampling errors, especially for the case of short time series, which do not reflect the effects of rare events. Alternatively these effects might be included with simulations of catastrophic (or unexpected) events (s.a. Heikkinen and Kuosmanen, 2002; Knoke et al., 2005; Knoke and Wurm, 2006).

Downside risk models are explicitly focused on the probability of falling below a predefined return level and show appeal for forest science. This is an important aspect and enables one to consider the risk factor as it is usually understood by many foresters, whose basic intention, among others, generally might not be profit maximization but in gaining a minimal acceptable return. Therefore, these approaches tend to produce slightly higher allocations to less risky assets than the MV approach (Grootveld and Hallerbach, 1999). The models allow clear differentiation of all alternatives due to the limited considered domain of the distribution function. However, even in this case, there still remain computational difficulties (Grootveld and Hallerbach, 1999) for which reason these models are not widely used for portfolio selection in practice (Bertsimas

et al., 2004). The danger of sampling errors seems to be limited, as only the left tail of the distribution function needs to be reliable, but as Grootveld and Hallerbach (1999) have shown, there also can result large estimation errors due to neglecting parts of the data set. Finally, as mentioned in the context of SD approach, the left tail of the distribution function is generally difficult to estimate due to low information about catastrophic events.

The approach that offers specific information about sensitivity to sampling errors or unexpected changes is the IGDT. This approach is slowly percolating into ecological research fields and its acceptance seems to be growing. Its high acceptance in ecological studies was the reason why we chose to devote so much space to introducing IGDT. However, IGDT has also provoked heated and emotional criticism. Sniedovich (2007) has launched a campaign against IGDT, not seeing the opportunity of IGDT to form a bridge that connects various disciplines. Sniedovich (2007) criticizes IGDT as being only a simple instance of Wald's maximin model and as local by its nature (Sniedovich, 2010). Naturally, that uncertainty set what is searched for by IGDT, i.e. that special set of parameters, which would still allow for acceptable outcomes, may only be a subset of the actual, but usually unknown uncertainty set. But nevertheless information about this subset is most welcome and informative to compare alternative decisions. However, the critique by Sniedovich (2010), although often presented overly harsh, should be taken seriously. It should be an indication to consider, besides IGDT, also the numerous existing approaches on robust optimisation (e.g. Goldfarb and Iyengar, 2003; Bertsimas and Sim, 2004; Ben-Tal et al., 2006a). However, these approaches are not always easy to adopt for non-mathematicians.

A principal assessment of the portfolio's optimisation robustness, relating to Knightian uncertainty however, includes the specific financial intention of the investor and is thus meaningful. Furthermore, it enables one to place the value of diversification into at least more detailed terms: Comparing the estimated probability of failure with the probability which is needed to still obtain a minimum required reward quantifies the difference of probabilities which might be caused by unexpected changes. This allows for assessment of risk reduction under the consideration of uncertainty. Doing so can be rather important due to the long production periods in forestry (especially with regard to decisions about future stand mixtures, as in this case, the decision remains almost unchangeable for remaining planning horizons) or other decisions under severe uncertainty. For this reason, the maximisation of financial robustness according to various theoretical approaches is an interesting challenge in decision-making. Techniques beside IGDT to consider ignorance in connection with predictions based on past data such as robust optimisation approaches (e.g. Goldfarb and Iyengar, 2003; Bertsimas and Sim, 2004; Ben-Tal et al., 2006a; Natarajan et al., 2009) or probability bounds analysis (Person and Ginzburg, 1996) should be tested for this reason as well.

All presented approaches can also serve to illustrate the effects of diversification in forest investment, which needs to be investigated more intensively. In conclusion, assessing diversification in ecosystem management from an economic point of view, which may be extended to land use problems (Knoke et al., 2009a) and ecosystem services (Knoke et al., 2009b), can be improved through more intensive investigation in general and into optimising the adjustment of employed approaches, using empirical examples pulled from different perspectives.

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